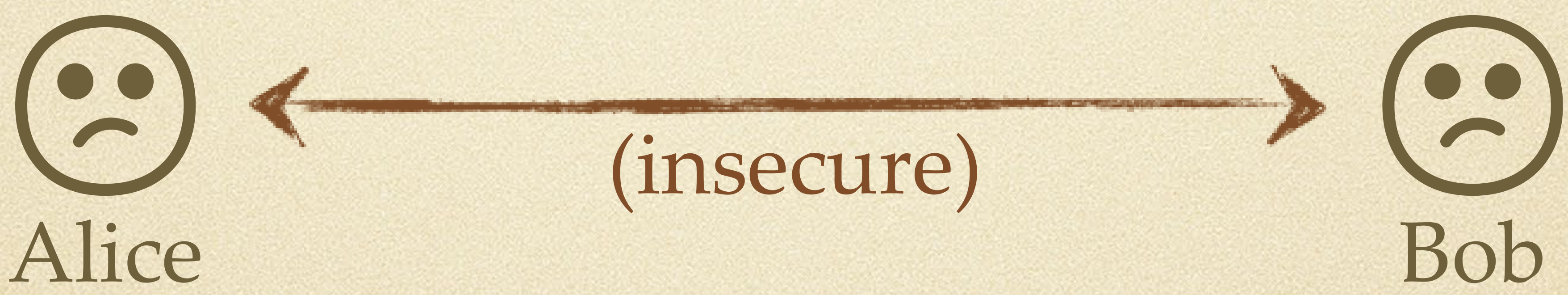
MATH3302/MATH7332

Diffie-Hellman Key Exchange
and
Elliptic Curve Cryptography

The Diffie-Hellman key exchange protocol

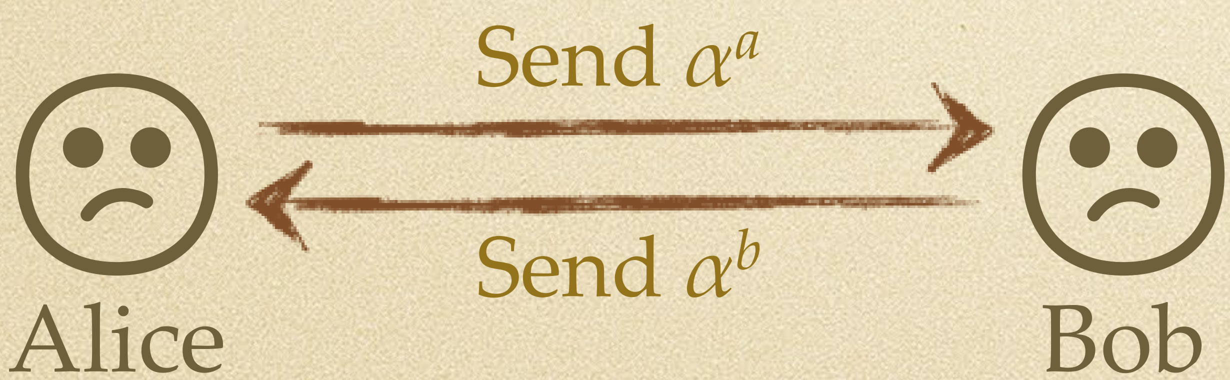
Aim:
To agree on a secret key over an insecure channel



Based on the difficulty of discrete logarithms (like ElGamal)

Diffie-Hellman details

- Alice and Bob publicly agree on a large prime p, and a generator α of $\mathbb{Z}_p^\times$
- Alice secretly chooses random exponent $a \in \mathbb{Z}_p^\times$ with $a \neq 1$ or $p-1$
- Bob secretly chooses random exponent $b \in \mathbb{Z}_p^\times$ with $b \neq 1$ or $p-1$



Final key:
$\alpha^{ab} = (\alpha^a)^b = (\alpha^b)^a$

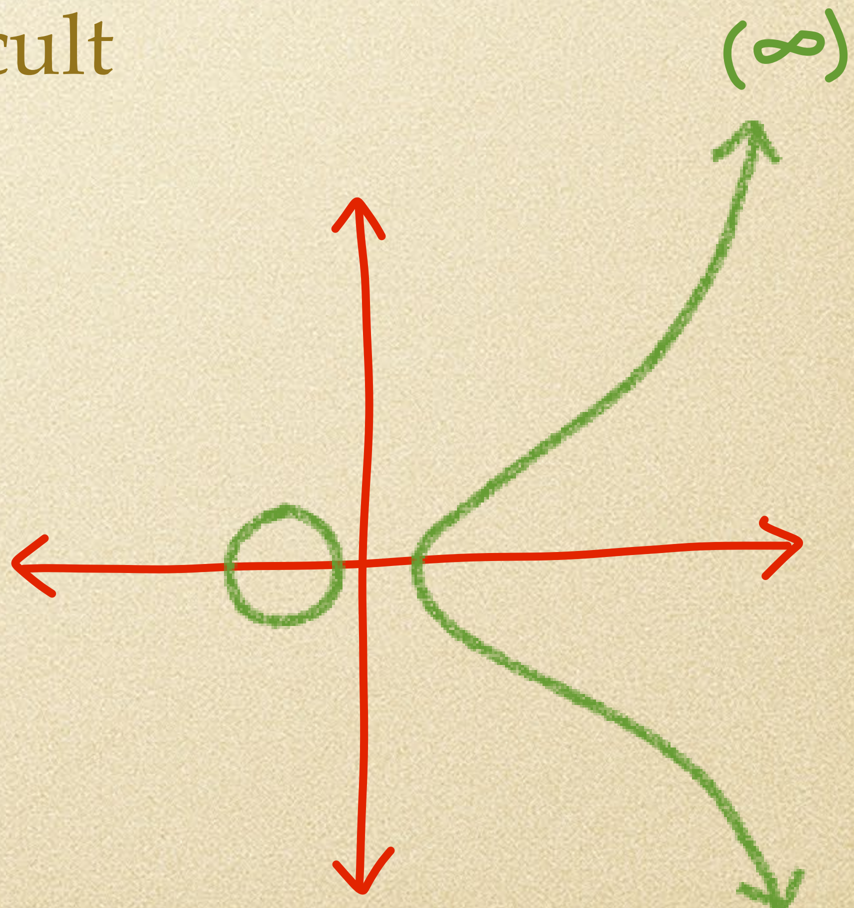
Elliptic Curve Cryptography

Why not use Diffie-Hellman and/or ElGamal with a different group?

- Any group will do, as long as discrete logarithms are still difficult

Elliptic curve : $y^2 = x^3 + ax + b$

Our group will be the points on an elliptic curve, along with the operation of "addition" on these points.



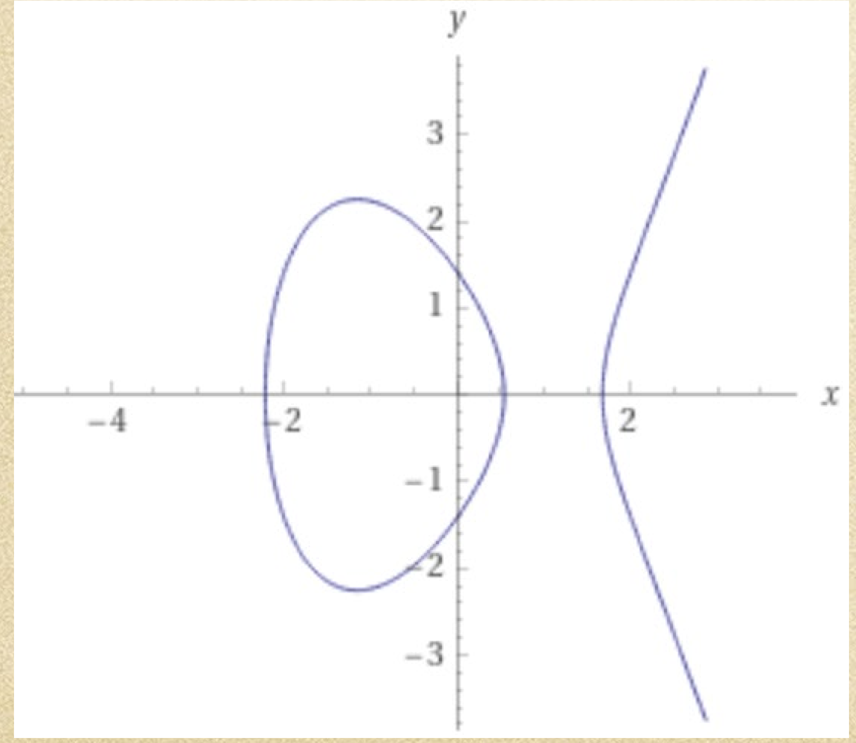
Elliptic Curves

Elliptic curves over the real numbers.

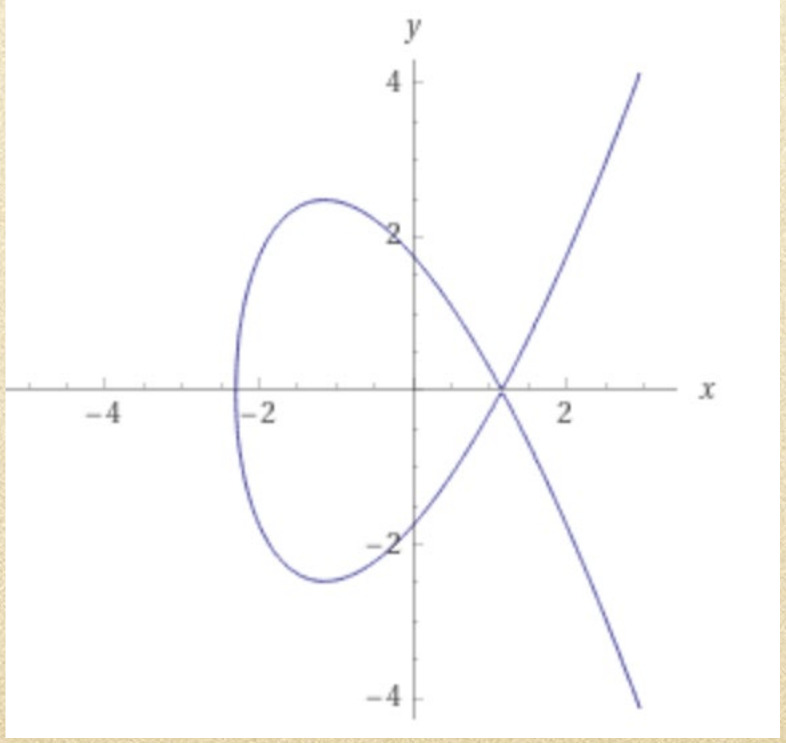
Elliptic curve : $y^2 = x^3 + ax + b$

A non-singular elliptic curve is one that has a well-defined tangent line at each point, this requires $4a^3 + 27b^2 \neq 0$.

$y^2 = x^3 - 4x + 2$



$y^2 = x^3 - 4x + \sqrt{\frac{4^4}{27}}$



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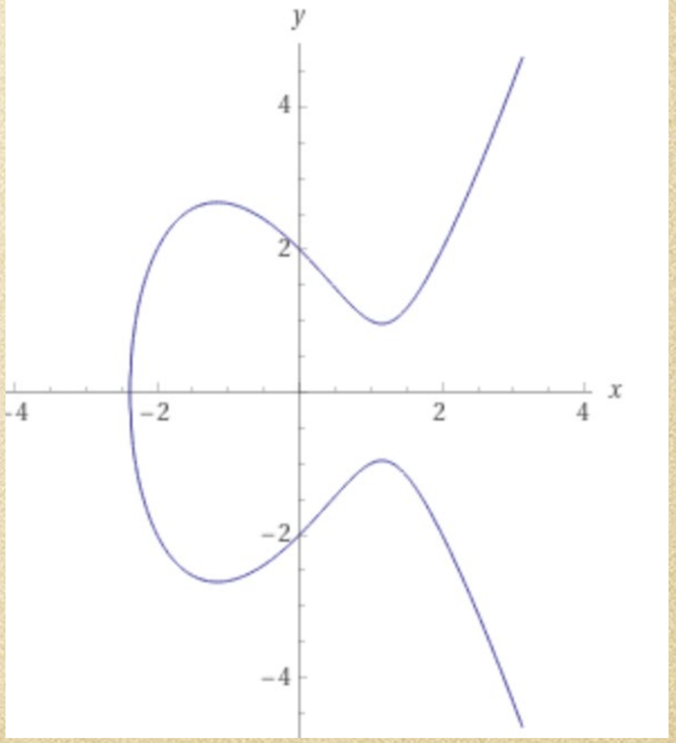
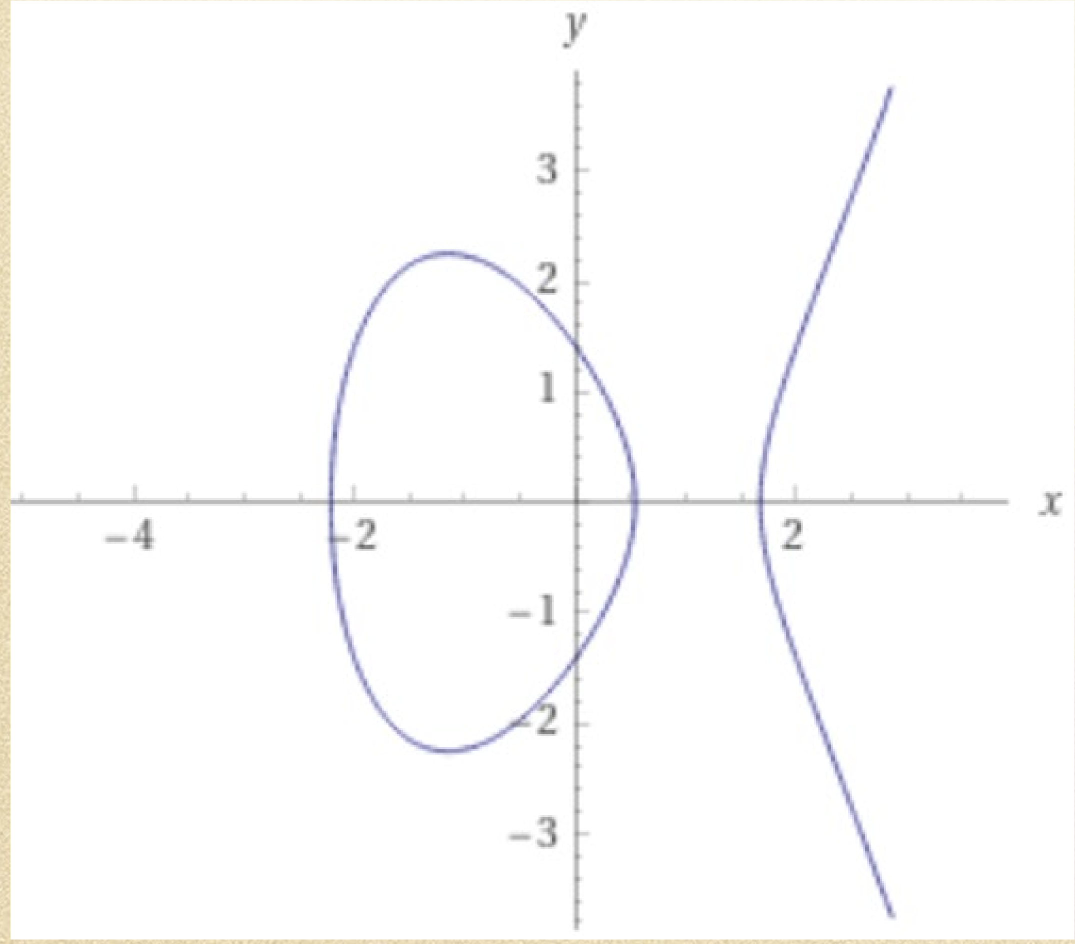
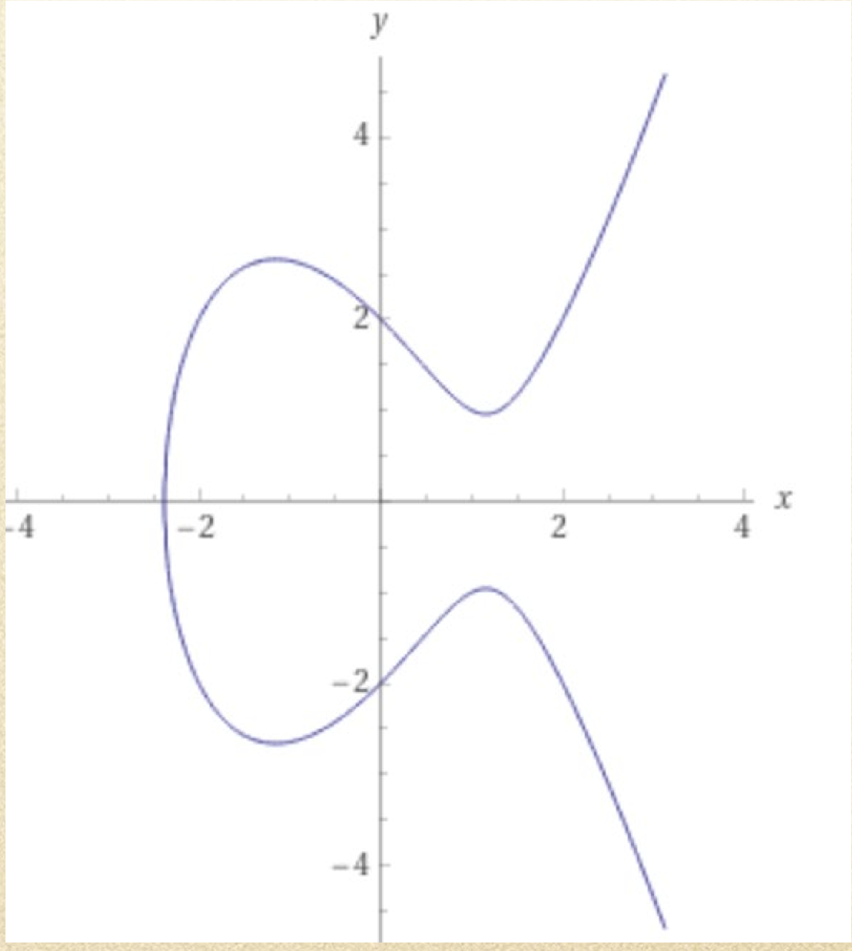


Elliptic Curves

Non-singular elliptic curve : $y^2 = x^3 + ax + b$, with $4a^3 + 27b^2 \neq 0$

Consider lines that intersect the curve at points P and Q.

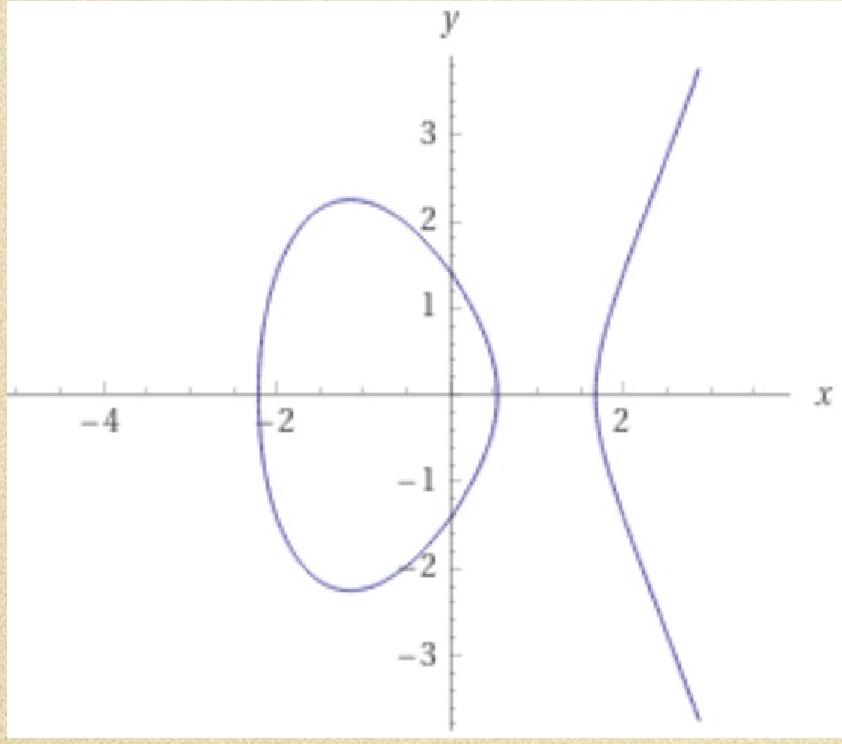


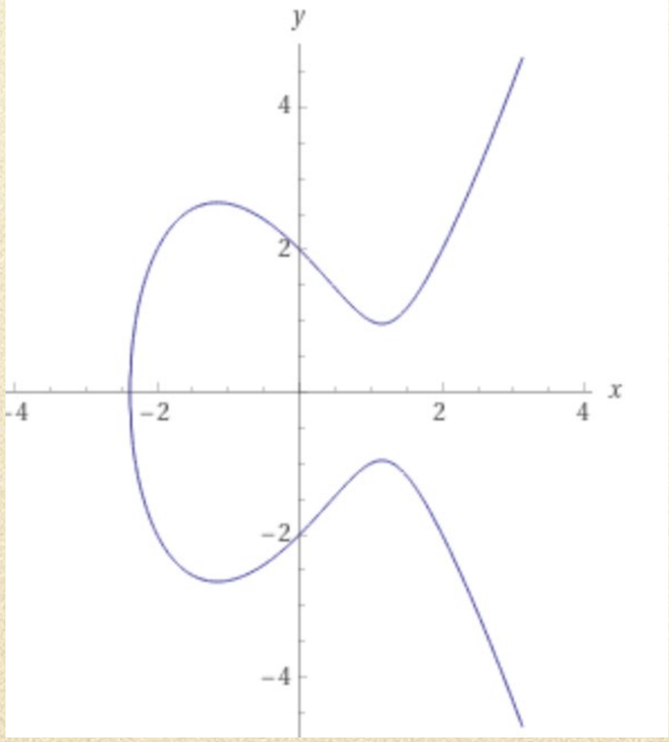


Elliptic Curves

Consider the set of points (x, y) that satisfy a non-singular elliptic curve $y^2 = x^3 + ax + b$, along with a point at infinity ∞^*.

To add points P and Q, draw a line through P and Q and let R be the "third" point in which that line meets the curve. Then reflect R in the x-axis to get the point $P + Q$.





Cases: (1) $P \neq Q$ and $P, Q \neq \infty^*$: use geometric description above

(2) $P = Q \neq \infty^*$: draw tangent at P for line in description above

(3) P or Q is ∞^*: use $P + \infty^* = P$, $\infty^* + Q = Q$, $\infty^* + \infty^* = \infty^*$

Elliptic Curves

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This set of points with the operation addition as defined above is an abelian group.
What is the identity?
What is the inverse of point $P = (x, y)$?

Elliptic Curves over $\mathbb{Z}_p$

Consider the set of points (x, y), with $x, y \in \mathbb{Z}_p$ for some prime p, that satisfy a non-singular elliptic curve $y^2 \equiv x^3 + ax + b \pmod{p}$, along with a point at infinity ∞^*.

Exercise: Determine all the points on the elliptic curve $y^2 \equiv x^3 + x + 6 \pmod{11}$.

Elliptic Curves over $\mathbb{Z}_p$

Useful fact regarding square roots modulo p:
If $p \equiv 3 \pmod 4$ is prime and $a \not\equiv 0 \pmod p$, then if a is a quadratic residue, then $\sqrt{a} \equiv \pm a^{\frac{p+1}{4}}\ (mod\ p)$.

This fact speeds up the search for points on elliptic curves modulo a prime $p \equiv 3 \pmod 4$.

Exercise: Consider the elliptic curve $y^2 \equiv x^3 - 2x + 4 \pmod{19}$.
(a) Show that there is no point on this curve having $x = 2$.
(b) Determine the two points on this curve having $x = 8$.

Elliptic Curves over $\mathbb{Z}_p$

Consider the set of points (x, y), with $x, y \in \mathbb{Z}_p$ for some prime p, that satisfy a non-singular elliptic curve $y^2 \equiv x^3 + ax + b \pmod{p}$, along with a point at infinity ∞^*.

Theorem (Hasse): If E is an elliptic curve over $\mathbb{Z}_p$, where $p > 3$ is a prime, then $|E| =$ the number of points in E satisfies

$$p + 1 - 2\sqrt{p} \leq |E| \leq p + 1 + 2\sqrt{p}.$$

Theorem (Waterhouse): If p > 3 is a prime and N is a positive integer that satisfies

$$p + 1 - 2\sqrt{p} \leq N \leq p + 1 + 2\sqrt{p},$$

then there exists a non-singular elliptic curve over $\mathbb{Z}_p$ with exactly N points.

Elliptic Curves over $\mathbb{Z}_p$

Example: In the group defined by the points on the elliptic curve $y^2 \equiv x^3 + x + 6 \pmod{11}$, determine the sum $(2,7) + (5,2)$.

Elliptic Curves over $\mathbb{Z}_p$

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Elliptic Curves over $\mathbb{Z}_p$

Addition of points on elliptic curves over $\mathbb{Z}_p$

Given: Points $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ on a non-singular elliptic curve E over $\mathbb{Z}_p$ defined by $y^2 = x^3 + ax + b \pmod{p}$ where $p > 3$ is prime.

If either P or Q is ∞^*, then $P + Q$ is the other point.

Otherwise, calculate

$$m = \begin{cases} (y_2 - y_1) \cdot (x_2 - x_1)^{-1} \pmod{p}, & \text{if } P \neq Q \\ (3x_1^2 + a) \cdot (2y_1)^{-1} \pmod{p}, & \text{if } P = Q \end{cases}$$

If m is undefined (inverses do not exist), then $P + Q = \infty^*$.

In all remaining cases, set $x_3 = m^2 - x_1 - x_2 \pmod{p}$ and $y_3 = m(x_1 - x_3) - y_1 \pmod{p}$, and then $P + Q = (x_3, y_3)$.

Elliptic Curves over $\mathbb{Z}_p$

The points of any non-singular elliptic curve over $\mathbb{Z}_p$ with the addition of points as defined on the previous slide is a finite abelian group.

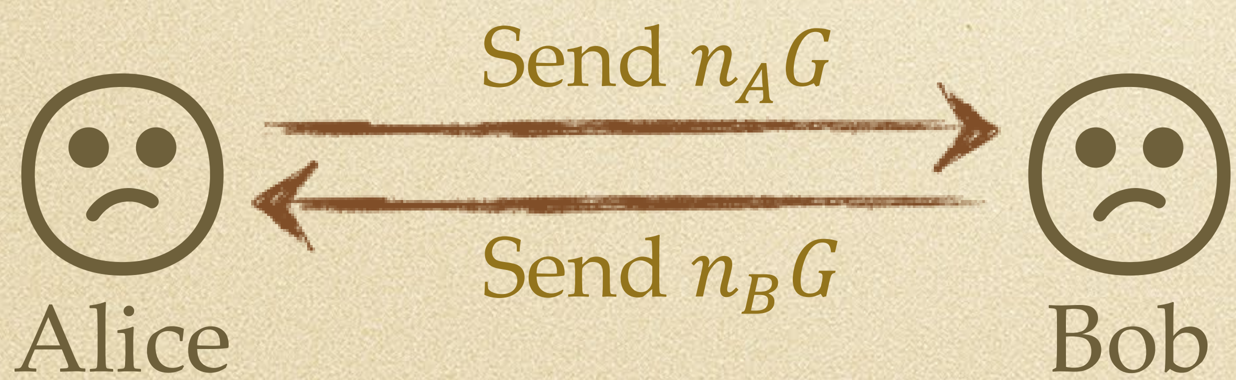
Discrete logarithm problem for $\mathbb{Z}_p^*$: Given a generator g of $\mathbb{Z}_p^*$ and given an integer $a \in \mathbb{Z}_p^*$, determine the exponent j for which $g^j \equiv a \pmod{p}$.

Discrete logarithm problem for elliptic curves over $\mathbb{Z}_p$:
Given an elliptic curve E over a prime $p > 3$, and points $P, Q \in E$ determine the positive integer m for which $Q = mP$, if such an integer exists.

Note that we want the point P to have high order.

Diffie-Hellman with Elliptic Curves

- Alice and Bob publicly agree on a large prime p, an elliptic curve E (mod p), and a point $G \in E$ (of high order)
- Alice secretly chooses random integer n_A such that $\sqrt{p} < n_A < p - \sqrt{p}$.
- Bob secretly chooses random integer n_B such that $\sqrt{p} < n_B < p - \sqrt{p}$.



Final key: x-coordinate of point $K \in E$ such that $K = n_A(n_B G) = n_B(n_A G)$.

How to compute nG

Given a large prime p, an elliptic curve E (mod p), and a point $G \in E$ (of high order), how do we calculate nG efficiently?

Use doubling!

Example: To compute $21G$, first calculate

- $2G = G + G$
- $4G = 2G + 2G$
- $8G = 4G + 4G$
- $16G = 8G + 8G$

Then $21G = 16G + 4G + G$.

Remember ElGamal?

Setup:

- Choose large prime p, generator α of $\mathbb{Z}_p^\times$, random $a \in \mathbb{Z}_p^\times$ with $a \neq 1$ or $p-1$
- Compute $\beta = \alpha^a \bmod p$
- Public key: (p, α, β)
- Private key: a

Encryption / decryption:

- $\mathcal{P} = \mathcal{C} = \mathbb{Z}_p$
- For each new plaintext, Alice chooses random $d \in \mathbb{Z}_p^\times$ with $d \neq 1$ or $p-1$
- $e_K(x) = (\alpha^d, \beta^d x) \bmod p$
- $d_K(c_1, c_2) = (c_1^a)^{-1} c_2 \bmod p$

Works entirely over the multiplicative group $\mathbb{Z}_p^\times$

ElGamal on an Elliptic Curve group

Setup:

- Choose large prime p, elliptic curve E mod p, point $G \in E$, random $n \in \mathbb{Z}$ with $\sqrt{p} < n < p - \sqrt{p}$
- Compute $A = nG$
- Public key: (p, E, G, A)
- Private key: n

Encryption / decryption:

- $\mathcal{P} = \mathcal{C} =$ points of E
- For each new plaintext P, Alice chooses random $n' \in \mathbb{Z}$ with $n' \neq 0$
- $e_K(P) = (n'G, n'A + P)$
- $d_K(c_1, c_2) = -nc_1 + c_2$

Works entirely over the elliptic curve group

$$d_K\big(e_K(P)\big) = d_K(n'G, n'A + P) = -n(n'G) + (n'A + P) = -n'(nG) + n'(nG) + P = P$$

Pros and cons

Advantages:

- Comparable security to RSA requires much smaller keys ⇒ good for hardware

Disadvantages:

- Plaintext space (points on the curve) is impractical to work with ... but see ECIES for a workaround
- Vulnerable to quantum computing
- Patent issues